

Hip Diagnosis and Decision-Making

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OVERVIEW OF PATHOLOGIES

Understanding the differential diagnosis for hip pathology is necessary before collecting a history and performing a physical examination. This background allows the clinician to tease out important elements in the history to narrow the differential and provide a focus for the physical examination. An overview of hip pathology is presented in the following sections (Box 77.1).

Soft Tissue Injuries

Inflammation and pain originating from the trochanteric, ischial, iliopsoas, and iliopectineal bursae are common. Movement at bone and soft tissue interfaces leads to repetitive friction and inflammation in these areas. Trochanteric bursitis is frequently diagnosed but may be more accurately described as greater trochanteric pain syndrome (GTPS). Magnetic resonance imaging (MRI) findings for this diagnosis have shown frequent abductor tendinitis and tears with bursitis lacking in up to 80% of patients.¹ Nonetheless, patients with trochanteric bursitis describe lateral thigh pain that can be reproduced with palpation on examination. Ischial bursitis typically presents with pain upon sitting and can be reproduced by palpation over the ischial tuberosity (IT). The iliopsoas bursa lies between the iliopsoas muscle and pelvic brim.² Patients typically present with inguinal pain that is reproducible with provocative maneuvers such as the Thomas test. The iliopectineal bursa is adjacent to the iliopsoas bursa but lies over the iliopectineal eminence. Symptoms are similar to those of iliopsoas bursitis, although iliopectineal bursitis can be seen in conjunction with a snapping iliopsoas muscle over the iliopectineal eminence.³

Snapping hip syndrome, also referred to as *coxa saltans*, causes an audible or palpable “snap” with hip range of motion (ROM). The etiology of a snapping hip is classified as external, internal, or intra-articular.⁴ An external snapping hip is caused by the iliotibial band (ITB) passing over the greater trochanter. An internal snapping hip is attributed to shifting of the iliopsoas tendon from medial to lateral over the femoral head, iliopectineal eminence, iliacus muscle, or lesser trochanter among others during hip flexion and extension.^{3,5-7} An intra-articular snapping hip is related to labral tears, loose bodies, or osteochondral injuries.

Contusions involving the hip, thigh, and pelvis are frequently encountered in athletes and occur after low-energy trauma. An iliac crest contusion, or “hip pointer,” results from a direct blow

to the iliac crest, and an overlying hematoma often develops. Quadriceps contusions typically involve a direct blow to the anterior thigh, which can result in hematoma formation and difficulty ambulating. A direct blow to the inner thigh may result in a groin contusion. Myositis ossificans can occur after a contusion and hematoma.⁸ The hematoma organizes and calcifies, which can lead to pain and stiffness. Myositis ossificans can also be seen in the absence of trauma.

Muscle strains and ligamentous injuries around the hip can be quite debilitating. Strains typically involve tearing at the musculotendinous junction and often occur during an eccentric contraction.⁹ Strains are classified by the affected muscle groups, including adductor, iliopsoas, external oblique, hamstring, and quadriceps. With significant force during athletics, or in the setting of trauma, the strong sacroiliac ligaments can be sprained. Pain typically originates in the lower back and radiates into the buttock or groin.

Hernias involve the extrusion of abdominal contents through a defect in the abdominal wall. A delay in diagnosis is common because hernias can mimic other conditions that cause groin pain with activity. Three hernias can present as hip or pelvic pain: inguinal, femoral, and sports hernias. Inguinal hernias involve the protrusion of abdominal contents through the deep inguinal ring or medial to the deep inguinal ring. Femoral hernias occur when a hernia sac protrudes through the femoral sheath to enter the anterior thigh. The “sports hernia” or athletic pubalgia is an increasingly recognized condition caused by posterior inguinal/abdominal wall weakness or tearing in athletes. No true protrusion of abdominal contents occurs, but patients experience chronic groin pain that is often difficult to diagnose.¹⁰ The pain can also radiate to the perineum and origin of the adductors. Sports that require frequent pivoting and cutting such as ice hockey, soccer, and American football have the highest incidence of this injury. Athletes often lack symptoms with rest but experience pain with activity or sport that prevents them from playing to their potential.¹¹

Bone Injuries

Sport-related trauma can involve significant energy resulting in fractures of the pelvis and femur. Pelvic ring injuries, acetabular fractures, femoral head and neck fractures, peritrochanteric fractures, and femoral shaft fractures lead to the acute onset of pain and difficulty or inability to weight bear or ambulate. Prompt recognition and treatment of fractures about the pelvis

Abstract

The evaluation of patients with hip pathology can be challenging. Due to the depth of the hip joint and numerous surrounding anatomic structures, patients often have difficulty describing the location and characteristics of their pain. This can produce a broad differential diagnosis for the treating physician. In addition, hip-related diagnoses frequently occur concomitantly with other local pathology, such as the spine. The goal of a thorough history should be able to rule out urgent sources of hip pathology, including malignancy, infection, or systemic disease; differentiate true hip pain from back pain; and narrow the differential diagnosis to perform a focused physical examination. A complete physical examination should include an evaluation of gait and stance, range of motion of bilateral hips in all planes, strength measurements throughout the entire lower extremity, multiple individual areas of palpation, and specialized maneuvers testing for specific pathology in the supine, lateral, and prone positions. An intra-articular diagnostic injection is an invaluable tool that helps rule out hip pathology with the clinical picture is unclear.

Keywords

hip
differential diagnosis
patient history
physical examination
hip-spine syndrome

BOX 77.1 Overview of Hip Pathology

Soft Tissue Injuries

Bursitis
Trochanteric
Ischial
Iliopsoas
Iliopectineal
Snapping hip syndrome
Contusions
Iliac crest
Quadriceps
Groin
Myositis ossificans
Strains
Adductor
Iliopsoas
External oblique
Hamstring
Quadriceps
Sacroiliac sprain
Hernias
Inguinal
Femoral
Sports (athletic pubalgia)

Bone Injuries

Traumatic fractures
Dislocation

Stress fractures
Pelvic
Sacral
Femoral neck
Osteitis pubis
Osteonecrosis

Degenerative Joint Disease
Nerve Entrapment Injuries

Sciatic
Obturator
Pudendal
Ilioinguinal
Femoral
Lateral femoral cutaneous

Intra-Articular Pathology

Labral tears
Femoral acetabular impingement
Loose bodies
Chondral injuries
Ruptured ligamentum teres

Infection

Pediatric Conditions

Avulsion fracture
Slipped capital femoral epiphysis
Legg-Calvé-Perthes disease

and femur are critical, as delay in diagnosis can result in long-term morbidity.

Stress fractures of the pelvis and femur occur in the setting of repetitive submaximal loading of bone. Pain that is aggravated by activity and subsides with rest is the hallmark feature of a stress fracture. Pelvic rami and sacral stress fractures are seen in athletes who participate in high-impact activities such as running and jogging. The pain is typically in the groin, buttock, or thigh when the ramus is involved and in the low back when the sacrum is the source.¹² Femoral neck stress fractures typically present with activity-related groin pain. The location of the stress fracture is critical to determining treatment. Tension-sided femoral neck stress fractures along the superior lateral neck require surgical treatment to prevent nonunion, avascular necrosis, or fracture displacement. Compression-sided femoral neck stress fractures occur along the inferior medial neck and are often amenable to nonoperative treatment.¹³

A stress injury of the pubic symphysis can lead osteitis pubis.¹⁴ This is frequently secondary to a muscular imbalance between the rectus abdominis (pelvic elevator) and the adductor tendons (pelvic depressor) that attach to fibrocartilage plate of the pubic symphysis. The pain is typically insidious in onset and is located at the midline over the symphysis or referred to the groin. Degeneration of the symphyseal cartilage can be followed by boney lytic changes, sclerosis, and widening of the symphysis.¹¹

Osteonecrosis, or avascular necrosis, of the femoral head is a cause of hip pain in young adults. Many conditions have been associated with osteonecrosis; however, the majority are related

to corticosteroid use, trauma, alcohol abuse, and coagulopathy.¹⁵ No cause is identified in 10% to 20% of cases, and this type of necrosis is termed “idiopathic avascular necrosis.” Patients typically present with pain in the groin or buttock and often walk with a limp. Bilateral avascular necrosis has been found in 40% to 80% of patients.^{15–17} Early identification may allow treatment that can prevent femoral head collapse and the need for arthroplasty.

Instability

Hip dislocations typically occur when significant force disrupts the soft tissue restraints of the hip joint.¹⁸ However, low-energy injuries may also cause hip dislocations in the setting of femoroacetabular impingement (FAI) morphology (cam, pincer, acetabular retroversion). Clinical and basic science studies support the mechanism of the anterior impingement levering the femoral head posteriorly.¹⁹ Hip dislocations are typically posterior, which leads to the patient having a shortened, internally rotated, and abducted hip. Less commonly, the dislocation will be anterior, which presents as an externally rotated and abducted hip. Expedient recognition and reduction may be essential to prevent avascular necrosis.

Microinstability of the hip is a more recent, but increasingly recognized, cause of hip pain. Several predisposing factors have been identified and include: generalized ligamentous laxity, repetitive microtrauma, ligamentum teres (LT) injuries, varying degrees of acetabular dysplasia, and iatrogenesis. Soft tissue and dynamic stabilizers of the hip such as the capsule, labrum, LT, and tendinous structures such as the iliopsoas may all play a role in microinstability. The clinical presentation may be subtle and difficult to detect. Most patients complain of pain in the groin, buttock, or thigh, but some may note “giving way” or apprehension. The onset is insidious with gradual worsening of symptoms.^{20,21}

Degenerative Joint Disease

Degenerative joint disease (DJD) results from the loss of cartilage in the hip joint and leads to progressive pain and stiffness. Many pathologic processes can lead to DJD, including but not limited to osteoarthritis, rheumatoid arthritis, avascular necrosis, infection, trauma, FAI, and dysplasia/instability. Although the process may be idiopathic, we now recognize that structural abnormalities of the hip are frequently associated with DJD.²² The diagnosis is usually confirmed by observing joint space narrowing, subchondral cysts and/or sclerosis, and/or osteophyte formation on plain radiographs. Patients typically present with the insidious onset of hip pain that worsens with activity.

DJD is a spectrum and ranges in severity. While advanced DJD is usually obvious, milder forms of DJD may be more ambiguous. It is crucial to determine how much joint degeneration is present, as this can have dramatic implications on treatment and surgical outcomes.²³ In addition to plain x-rays, delayed gadolinium-enhanced magnetic resonance imaging of cartilage (dGEMRIC) can provide a helpful quantitative analysis of the articular cartilage health.²⁴ An algorithm (Fig. 77.1) helps stratify patients by degree of arthritis and the appropriate respective treatments.²³

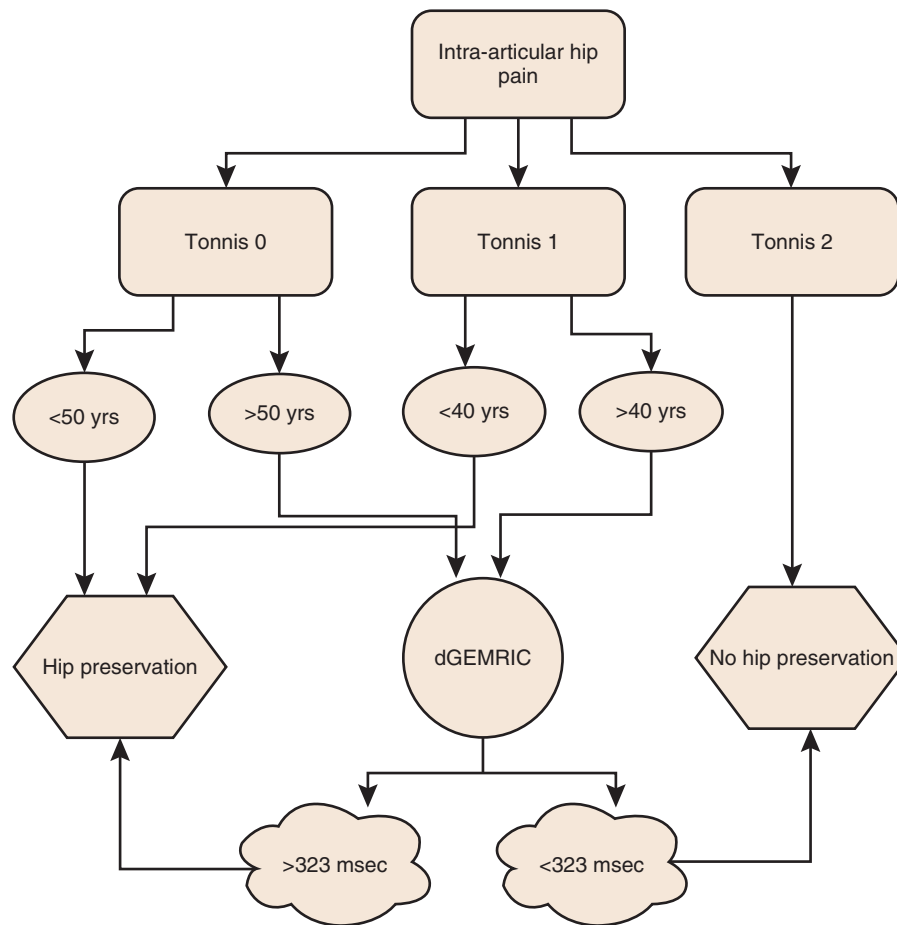


Fig. 77.1 Algorithm for patient selection for hip arthroscopy using radiographic and delayed gadolinium-enhanced magnetic resonance imaging of cartilage (*dGEMRIC*) parameters as a guide.

Nerve Entrapment Injuries

Nerve entrapment surrounding the hip can involve the sciatic, obturator, pudendal, ilioinguinal, femoral, and lateral femoral cutaneous nerve. Diagnosing and treating these conditions can be difficult and frustrating for the patient and clinician. The pain often has a burning quality and is confined to a nerve root distribution. Electromyographic nerve conduction studies are helpful in confirming the diagnosis and ruling out a lumbar radiculopathy. Sciatic nerve entrapment often presents with pain radiating down the buttock and posterior thigh. In some cases, this pain may be related to piriformis syndrome.²⁵ Obturator nerve entrapment causes pain in the medial thigh that can radiate toward the knee.²⁶ Prolonged compression during activities such as cycling can cause pudendal nerve entrapment.²⁷ Numbness and pain in the perineum and shaft of the penis are typical. Ilioinguinal nerve entrapment is a very rare cause of inguinal pain that often radiates into the groin. Pain over the anterior thigh can be caused by femoral nerve entrapment, and when severe, it may cause quadriceps weakness and difficulty with gait.²⁸ Lateral femoral cutaneous nerve entrapment, or *meralgia paresthetica*, causes anterolateral thigh pain and numbness that extends toward the lateral knee.²⁹

Intra-Articular Pathology

Our understanding of intra-articular pathology has significantly expanded during the past few decades. Structural abnormalities of the hip joint seen in persons with dysplasia and FAI often lead to injury of the labrum and chondral surface. Hip dysplasia, or developmental dysplasia of the hip, results in a broad spectrum of disease. The underlying abnormality is inadequate acetabular coverage of the femoral head, which in severe forms can cause hip dislocations in children. Often the degree of undercoverage is mild or “borderline” and leads to pathology during adulthood because of the concentration of forces on a shallow acetabulum.³⁰ FAI is caused by abnormalities of the femur and acetabulum that lead to abnormal contact within the hip joint.³¹ Deformity on the femoral side is termed *cam impingement* and is due to an aspherical femoral head commonly located at the anterosuperior head-neck junction. Deformity on the acetabular side is termed *pincer impingement* and occurs with excessive acetabular coverage of the femoral head. Most commonly, both abnormalities exist in the same hip joint, and this condition is termed *combined impingement*. Iliopsoas impingement is compression and damage to the underlying anterior acetabular labrum at the 3 o’clock position (right hip) by a tight iliopsoas tendon or an inflamed tendon adherent to the anterior capsulolabral complex

causing a repetitive traction injury to the labrum. This differs from the typical 1 to 2 o'clock labral tear distribution seen in FAI.³² Patients with intra-articular pathology typically present with groin, lateral hip, or buttock pain. Scrutinizing hip radiographs will often help the clinician identify morphologic abnormalities of the hip that may lead to intra-articular pathology.¹⁴ Multiple other sources of intra-articular pathology have also been described, such as ruptured LT, loose bodies, and synovial disease.

Infection

Septic arthritis of the hip should always be considered in a patient who presents with the acute onset of pain. The patient is often febrile and lacks a history of trauma. Physical examination reveals pain with attempted passive ROM. Inflammatory markers are typically elevated. Prompt recognition and treatment are necessary to prevent long-term complications.³³

Pediatric Hip Conditions

Skeletally immature patients often present with hip conditions that differ from the hip conditions of adults. Open physes and apophyses are areas of weakness and are frequently injured. When muscles are overloaded in children, failure can occur at the origin of the muscle, particularly when an apophysis is present.³⁴ This scenario causes an avulsion fracture at the muscle origin, which differs from the pathology seen in adults, who most frequently experience a soft tissue tear of involving the tendon. Males account for up to 86.7% of these injuries. Avulsion fractures occur most commonly at the anterior superior iliac spine (ASIS), anterior inferior iliac spine (AIIS), and IT, when the sartorius/tensor fasciae latae, rectus femoris, hamstrings, and iliopsoas muscles, respectively, are overloaded. Lateral and distal displacement of ASIS injuries can be misdiagnosed as AIIS avulsions. Patients with IT avulsions tend to be younger due to earlier ossification center formation and 12% will have a contralateral sided injury. Less common avulsion injuries include the iliac crest, pubic symphysis, and lesser trochanter with muscle forces from the abdominal muscles, adductors, and iliopsoas, respectively. If misdiagnosed or delayed in presentation, these injuries can lead to hip impingement or nerve irritation and exhibit the respective symptoms.³⁵

Slipped capital femoral epiphysis (SCFE) is a disorder of the proximal femoral physis. The proximal femoral physis fails, leading to anterior superior displacement of the femur relative to the epiphysis.³⁶ This condition typically involves patients 11 to 14 years of age, often affects obese children, and may be seen in the setting of endocrine abnormalities. Displacement at the physis can frequently be identified with a frog-leg lateral radiograph. Duration of symptoms and stability at the physis is important for treatment and prognosis. Patients with an unstable slip are unable to weight bear and have approximately a 50% chance of AVN.³⁷ When SCFE is identified, surgical treatment is often indicated.

Legg-Calvé-Perthes disease (LCPD) is a childhood disorder that leads to ischemic necrosis of the growing femoral head. The process typically affects patients 5 to 8 years of age and predominantly involves boys. Like SCFE, there may also be an

association with obesity.³⁸ Parents notice that the child is limping, but the patient often has only mild pain. Radiographs can identify abnormalities in the femoral epiphysis.

HISTORY

The goal of the history should be to (1) rule out any alarming sources of hip pathology, including malignancy, infection, or systemic disease; (2) differentiate true hip pain from back pain; and (3) narrow the differential diagnosis to perform a focused physical examination. Patient age and profession or sport, chief complaint, history, and mechanism of trauma—including previous hip dislocations, developmental or childhood hip-related conditions, and previously attempted treatment modalities should all be ascertained. These modalities often include analgesic or antiinflammatory medications, assistive devices, interaction with other physicians, injections, physical therapy, or previous surgery.³⁹ A history of acute onset or trauma is a better prognostic indicator than an insidious onset of symptoms.⁴⁰ Important pain characteristics include: location, intensity, quality, onset, duration, radiation, alleviating factors, aggravating factors, and associated factors (Fig. 77.2). Mechanical symptoms such as snapping, popping, locking, clicking, and subjective instability should also be elicited. Pain or catching during flexion and axial loading activities (e.g., rising from a seated position, walking up or down stairs, or entering and exiting a vehicle) also implies mechanical hip pain.⁴¹ Patients may also admit to pain with sexual intercourse or difficulty putting on shoes or socks because of rotational force and complex hip movements.⁴²

The remainder of the standard history should not be overlooked. A medical and surgical history may uncover recent or

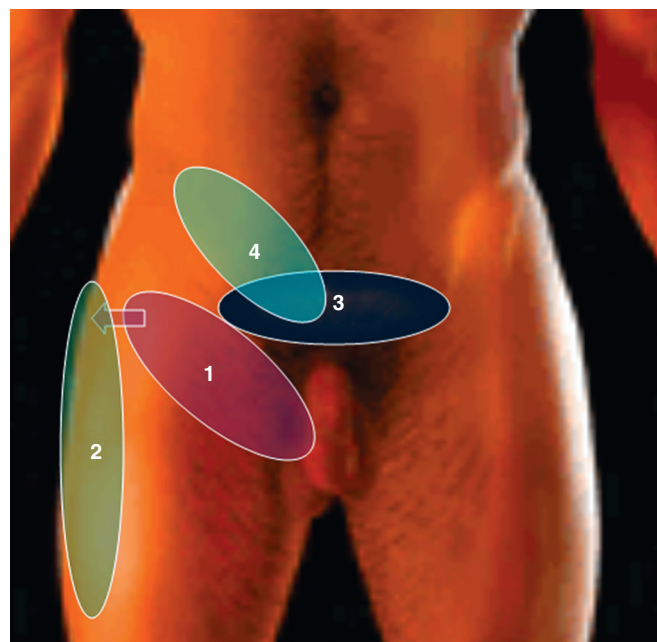


Fig. 77.2 The location of pain can aid in diagnosis. 1, Intra-articular pain can be caused by osteoarthritis or labral pathology. 2, Lateral hip pain can be caused by trochanteric bursitis or abductor contracture. 3, Symphyseal pain can be caused by osteitis pubis. 4, Abdominal pain can be caused by athletic hernias.

remote illness including malignancy, an immunocompromised state, tuberculosis or other infection, deep venous thrombosis, or hernia. A family history of cancer, systemic inflammatory arthritis, hip instability, or DJD is also useful. A social history detailing abuse of tobacco, alcohol, or illicit drugs is also important and may increase the index of suspicion for osteonecrosis of the femoral head or malignancy. Obtaining a sexual history as part of the social history can help evaluate the patient's risk of a sexually transmitted infection and associated septic arthritis. Pain with menses also should be noted. The social history should also include a review of the patient's activities of daily living, work activities, hobbies, athletic activities, and impairments to these activities that are a result of the chief complaint. Participation in running, basketball, soccer, ballet, hockey, golf, tennis, martial arts, or rugby is specifically associated with hip pathology.^{43–45} A review of systems including recent fever, chills, malaise, night sweats, or unintended weight loss warrants further investigation.⁴⁶ The clinician should remember that genitourinary—including pelvic floor musculature, gastrointestinal, neurologic, or vascular pathology can also masquerade as hip pain.

When eliciting the history, the clinician should ask the patient what he or she believes is causing the pain. The patient's expectations of the encounter should be addressed, and he or she should be given a specific opportunity to ask questions or raise concerns. In addition, the patient may provide or recall more pertinent history during the physical examination, and thus the clinician must always listen attentively (Box 77.2).

PHYSICAL EXAMINATION

The physical examination of the hip consists of several steps, including inspection, symptom localization, measurements, ROM, and special maneuvers.

Inspection

Observation of stance and gait inspection are essential components of the physical examination. The patient's stance should be inspected for a flexed position of the hip or knee, which may be indicative of hip pathology. The patient's stance should also be inspected for gross atrophy, spinal malalignment, or pelvic obliquity. While sitting, the patient should be observed for leaning to the unaffected side to avoid excessive flexion. The gait should be observed in multiple planes and for six to eight strides in the frontal and sagittal plane.⁴⁷

Patients may present with one of several gait abnormalities, including an antalgic gait, a pelvic wink, a Trendelenburg gait, excessive pelvic internal or external rotation, or true or false leg length discrepancies.³⁹ The antalgic gait involves a limp to decrease the time of stance phase on the affected side and limit weight bearing through the affected hip. An antalgic gait could indicate pain anywhere along the affected lower extremity. A pelvic wink is greater than 40 degrees of rotation in the axial plane toward the affected hip during terminal extension and signifies a hip flexion contracture in the setting of lumbar lordosis or forward-stooping posture. A Trendelenburg gait, also known as an abductor lurch, involves leaning the trunk toward the affected hip. Abductors normally stabilize the pelvis when the contralateral leg is raised. If the abductor muscles or the nerves supplying those muscles are injured or not functioning, the patient will lurch to the ipsilateral side to prevent pelvic sagging. A Trendelenburg sign can be elicited by having the patient raise the contralateral leg. A pelvic sag of more than 2 cm on the ipsilateral side is a positive Trendelenburg sign, indicating abductor pathology. Internal or external rotation should be noted both during gait and in the sitting position, which stabilizes the pelvis. Decreased internal rotation is a sign of internal hip pathology.⁴⁸ Excessive internal rotation with decreased external rotation indicates increased femoral anteversion.⁴⁹ A short leg limp during gait may indicate a true or false leg length discrepancy, or the limp may be secondary to ITB pathology. Children with leg length discrepancy may present with a circumduction gait or a vaulting gait to clear the long leg.⁵⁰

Snapping or clicking noises should be noted as the patient walks. These noises can indicate iliopsoas, ITB, or intra-articular pathology. Lateral snapping is more likely with ITB pathology and generally is easily visualized.⁵¹ Audible anterior snapping is more likely a result of iliopsoas tendon pathology. Either etiology of snapping may be palpable and/or reproducible.

Symptom Localization

Numerous techniques can help localize symptoms. A simple tool involves asking the patient to localize his or her pain by using a single finger. This technique can narrow the differential. In addition, examination of this location can be reserved for the end of the examination, thus enhancing the patient's trust by not causing pain at the beginning of the examination. This pain location in conjunction with Hilton's law can help distinguish the cause of pain in the hip region. Hilton's law states that "the same trunks of nerves whose branches supply the groups of muscles moving a joint furnish also a distribution of nerves to the skin over the insertion of the same muscles, and the interior of the joint receives its nerves from the same source."⁵² This law helps explain why muscle spasms and superficial pain accompany hip pathology. The L3 nerve predominantly innervates the hip joint, and thus pain down the entire L3 dermatome (the anterior and medial thigh crossing laterally to the knee) can accompany hip pain. This pain can be confused with compression of the lateral femoral cutaneous nerve, also known as meralgia paresthetica, which will cause pain or neuralgia in the L2 or L3 dermatome.^{53–56}

Another useful observation is the C sign. The C sign occurs when a patient cups his or her hand above the greater trochanter

BOX 77.2 Information Obtained From Patient History

Character and location of pain
 Mechanism of injury (be specific if possible)
 Duration of symptoms
 Activity-related pain (e.g., Does it subside with rest?)
 Pain related to bowel or bladder activity or ingestion of food
 Pain related to menses
 Treatment history (injections, physical therapy, other physician evaluations)



Fig. 77.3 The C sign. Cupping the thumb and the index finger around the lateral thigh indicates deep hip pathology.

when describing hip pain. The hand forms a “C” and is placed on the lateral hip, which may lead the clinician to mistakenly consider lateral hip pathology such as ITB or trochanteric bursitis (Fig. 77.3). However, this sign is most characteristic of deep interior hip pain.^{42,57} When a C sign is observed, the clinician should ensure that the examination and subsequent testing include a complete evaluation for intra-articular hip pain.

Palpation can also assist in localization of symptoms in the hip, although not as much as in other joints. The clinician must be familiar with the superficial and deep anatomy of the hip region for palpation to be of assistance. Initial palpation of the abdomen to check for fascial hernias or masses is important to rule out gastrointestinal or genitourinary sources of pain. Palpation of the lumbar spine, sacroiliac joint, ischium, iliac crest, greater trochanter, trochanteric bursa, various muscle bellies, rectus femoris and iliopsoas tendons, the adductor tubercle, and the pubic symphysis can help distinguish the location of pain. Pubic symphysis pain can indicate fracture, calcification, or osteitis pubis. The clinician can also palpate the femoral nerve. A positive Tinel sign indicates a neurologic cause of pain. Identification of bursitis on examination is often extremely helpful in narrowing the differential and can be followed up with an injection of a local anesthetic to confirm the diagnosis. The addition of a steroid to the injection is often therapeutic as well.

A distal neurovascular examination should also be completed, including palpation of the dorsalis pedis and posterior tibial arteries, an assessment of capillary refill, and lymphatic return. A thorough motor and sensory examination can often be helpful in differentiating hip from lumbar pain. Strength testing of the iliopsoas, quadriceps, extensor hallucis longus, tibialis anterior, and gastrocnemius provides an assessment of the L2-S1 nerve roots. Any asymmetric weakness compared with the contralateral side may indicate a radicular component to the pain, and this

finding should prompt further investigation into lumbar spine or peripheral nerve pathology. A thorough sensory examination will often help confirm radicular or peripheral nerve problems. When assessing numbness, the examiner should differentiate a radicular (L2-S1) pattern from a peripheral nerve pattern (femoral, peroneal) to provide a focus for additional testing. Intra-articular hip pain should not cause sensory disturbances or weakness in the foot. In the setting of injury, muscle testing also can be helpful. If the clinician believes that a specific muscle group has been affected, then resisted contraction can reproduce the patient's symptoms. Specifically, leg abductors (superior gluteal nerve L4-S1), leg adductors (obturator nerve L2-L4), knee extensors and hip flexors (femoral nerve L2-L4), hip extensors (inferior gluteal nerve L5-S2), and knee flexors and lower leg muscles (sciatic L4-S3) should be tested. The strength of patients with intra-articular pathology is often limited by pain, and the clinician should be aware that weakness in the iliopsoas or abductors may be a result of pain instead of being neurogenic. The algorithm in Fig. 77.4 is useful in the setting of hip-spine syndrome and an ambiguous diagnosis.⁵⁸ Diagnostic injections of the hip joint are invaluable in differentiating hip and lumbar pathology.^{59,60} If possible, the patient should perform a symptom-aggravating activity, be examined prior to the injection, observed while the anesthetic takes effect, and finally, re-examined. Differences in pain scale, ROM, and special maneuvers should be documented. Injection of local anesthetic concurrently with an MR arthrogram is unreliable, as the contrast agent itself can be irritating.

Active ROM, passive ROM, and resisted ROM may reproduce hip pain. ROM examination can help differentiate muscle strains from intra-articular pathology. In general, passive ROM may reproduce hip pain but not pain from a muscle strain. A strain of the hip flexors can be detected with the active hip flexion but not passive hip flexion.

Measurements and Range of Motion

Several measurements are important during the hip evaluation. Ipsilateral measurements from the shoulder of the iliac crest or from the ASIS to the medial malleolus can detect a true leg length discrepancy. Degenerative changes and collapsed avascular necrosis frequently cause a leg length discrepancy. Preoperative templating and the physical examination are the keys to successfully restoring leg length at the time of reconstruction.

The measurement of thigh circumference can detect muscle atrophy from chronic conditions. It is important to choose the same level of the thigh for circumference measurements. For example, 10 cm above the patella is a typical location for this measurement. The contralateral side must be measured for comparison, and this measurement can be taken at subsequent visits to gauge response to treatment.

Hip ROM must be accurately and reproducibly measured during the physical examination of the hip (Table 77.1). ROM can indicate the extent of hip pathology and the response to treatment. Hip flexion is best tested in the supine position, and the normal range is up to 120 degrees.⁶¹ Hip extension is best tested in the prone position, and the normal range is up to 15 degrees. Internal and external rotation should be tested with the

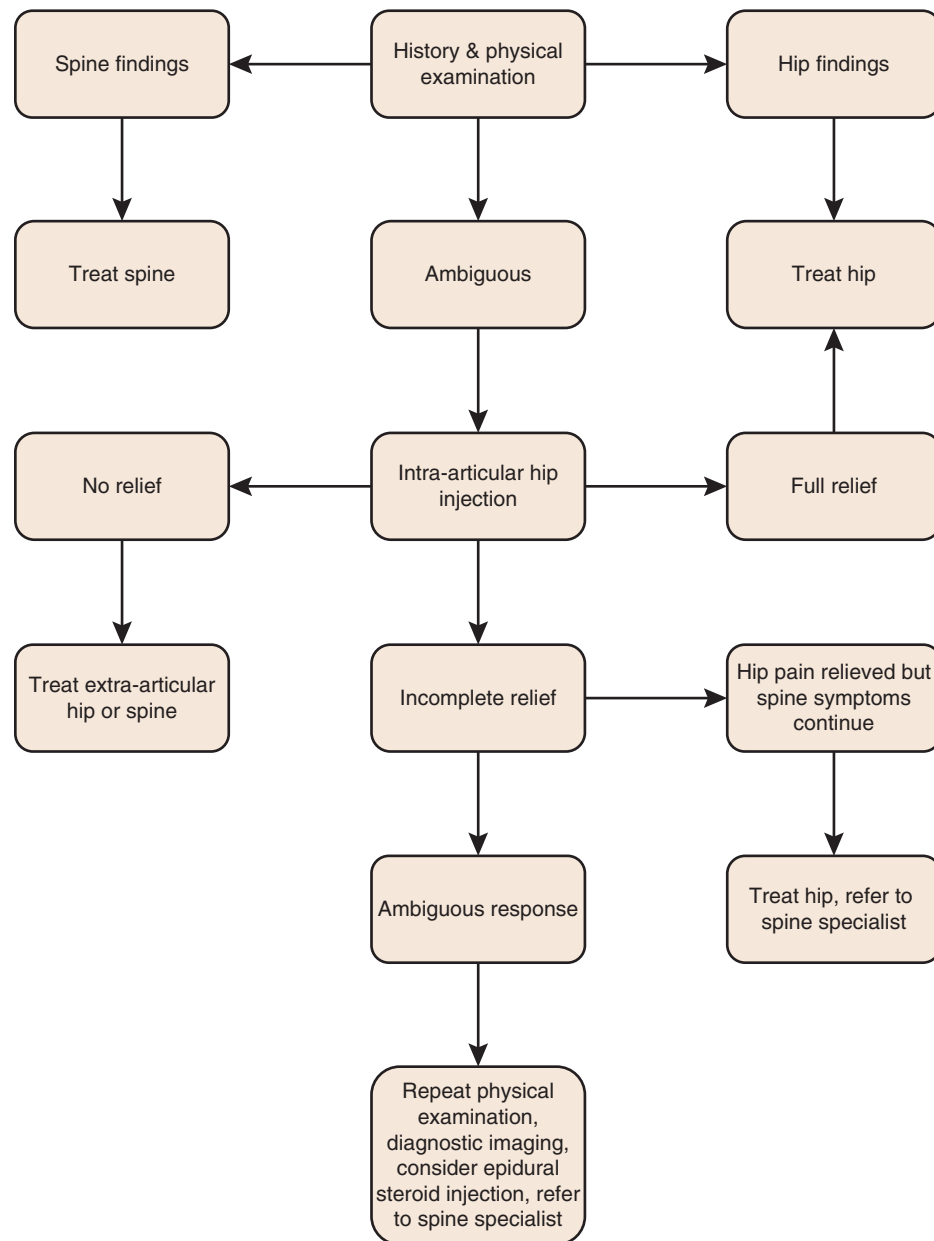


Fig. 77.4 Algorithm helping to differentiate hip and spine pain sources.

TABLE 77.1 Average Hip Range of Motion

Motion	Degrees
Flexion	115
Abduction	50
Adduction	30
Internal rotation	45
External rotation	45

hip flexed at 90 degrees, which is best accomplished in the seated position to stabilize the pelvis. Normal internal rotation is up to 45 degrees, and normal external rotation is up to 50 degrees.⁶¹ Arthritis, effusion, SCFE, FAI, or muscle contractures can cause a decrease in internal rotation.^{49,62} Abduction and adduction

should be measured with reference to the shaft of the femur at the midline of the pelvis. Normal abduction is 45 degrees, and normal adduction is up to 30 degrees.⁶¹ An abductor contracture may decrease adduction ROM.⁶¹ Differences between the normal and affected side can help detect hip pathology.⁴⁸ In patients with DJD, hip ROM is often globally decreased, with pain at the extremes of motion. Patients with dysplasia have shallow acetabuli and anterior superior undercoverage, which allows increased internal rotation with the hip flexed. Patients with FAI typically lack internal rotation in a flexed position because of the abnormal contact between the femoral head and neck with the acetabulum. After acetabular rim trimming and femoral osteoplasty, intraoperative ROM can be tested. Typically, as the impingement is relieved, the patient will gain internal rotation with the hip flexed.

Evaluation of Ligamentous Laxity

Testing for general ligamentous laxity is a good complement to the abduction–extension–external rotation examination when instability is suspected. Generalized laxity can be estimated by utilizing the Beighton criteria.⁷³ This gives the patient a numerical score out of a possible nine points, with one point assigned to each of the following with each the right and left sides accounting for unique scores: small finger passive dorsiflexion greater than 90 degrees, thumb passive dorsiflexion to the flexor aspect of the forearm, elbow hyperextension greater than 10 degrees, knee hyperextension greater than 10 degrees, and palms flat on the floor with forward flexion and knees fully extended. A score of four or greater denotes generalized ligamentous laxity and a possible concern for an underlying connective tissue disorder. A referral for medical and/or genetic evaluation may be warranted in some patients.⁴⁵

Special Maneuvers

Supine

The straight-leg raise (SLR) is a test to assess lumbar nerve root pathology. With the knee held in extension and the contralateral pelvis stabilized, the clinician passively raises the leg. When the patient feels pain, the hip should be extended 10 degrees and the foot dorsiflexed to elicit radicular symptoms. Pain from 0 to 30 degrees suggests a compressed nerve root, pain from 30 to 60 degrees may suggest sacroiliac disease, and pain after 60 degrees suggests sacroiliac disease.^{63,64} The pain should be similar to the patient's chief complaint rather than generalized stiffness. Stretching of the hamstrings should not be confused with a positive SLR test.

An SLR against resistance, also known as an active SLR, can indicate hip pain. Again, in the supine position, the patient flexes at the hip with the knee extended, but the clinician resists the flexion, thus generating force across the hip joint. Pain with the active SLR indicates hip pain.

The log roll test is the most specific test for intra-articular hip disease. The femur is gently rotated both internally and externally without motion of the pelvis or undue pressure on the surrounding muscles. Pain is specific but not sensitive for hip pathology. The heel strike and Stinchfield tests can be useful accompaniments to the log roll test. Striking the heel with the fist causes an axial load through the hip and can recreate hip pain. The Stinchfield test involves gradually increasing resisted flexion of the hip, with pain indicating either joint pain or iliopsoas pathology.⁴⁹ Plain radiographs or MRI will often confirm physical examination findings for intra-articular pain. If the history and physical examination findings are confusing, a diagnostic injection of local anesthetic in the hip joint can be helpful in ruling out extra-articular pain.

The anterior impingement test involves forced flexion, adduction, and internal rotation of the hip (FADIR). This test is the most sensitive maneuver to detect subtle hip pathology, including FAI. Because this maneuver may be uncomfortable in a normal hip, a comparison to the unaffected side is extremely important. This position of the hip loads the anterior superior aspect of the hip joint by bringing the femoral head and neck into contact

with the anterosuperior acetabulum. When this test is positive, the clinician should scrutinize the patient's radiographs for signs of FAI or DJD. If no degenerative changes are observed, further workup with an MRI arthrogram may be helpful to further evaluate the hip joint. A positive anterior impingement test may also guide decision-making intraoperatively by leading the surgeon to treat anterior pathology identified during arthroscopy.

The lateral impingement test is performed with the hip in extension and abducted. The knee is then flexed and the hip is internally rotated. A lateral cam on the femoral head/neck will impinge the superior aspect of the acetabulum causing pain and a positive test. The Patrick test, also known as the flexion, abduction, and external rotation (FABER) test, can stress the sacroiliac joint or imply lateral impingement or iliopsoas pathology. A figure-of-four is made in the supine position by placing the ipsilateral ankle over the contralateral knee. The ipsilateral knee is then pressed down. Posterior pain indicates sacroiliac joint disease, lateral pain indicates lateral impingement, and groin pain indicates iliopsoas pathology.⁴⁹

The McCarthy test involves bringing both hips in to full flexion in the supine position and then extending the affected hip in external rotation and internal rotation. Reproduction of the original pain during this arc of motion constitutes a positive McCarthy test and indicates labral pathology.^{65–69}

The Scour test can help detect irregularities along the femoral head or acetabulum. The clinician flexes the hip and knee while pointing the knee to the shoulder. While taking the hip through its arc of motion, any bumps or catches should be noted. The presence of bumps or catches indicates a positive test and is suggestive of FAI.

The Thomas test determines the presence and degree of a flexion contracture (iliopsoas contracture) of the hip. The patient should flex the unaffected hip and draw the knee to the chest while lowering the affected leg to the table. A positive Thomas test occurs when the patient cannot lower the affected extremity to the table. Any popping or clicking during the Thomas test may indicate a labral tear.⁴⁶

The Dial test of the hip can help detect hip capsular laxity. In the supine position, the patient's hip and knee are extended. The clinician places a hand on the femur and a hand on the tibia and internally rotates the entire lower extremity at the hip joint. The lower extremity is then released and allowed to externally rotate. External rotation less than 45 degrees with a firm end point constitutes a negative Dial test. Passive external rotation beyond 45 degrees constitutes a positive Dial test and is indicative of hip capsular laxity.⁴⁷

The posterior apprehension test can detect posterior hip micro or macro instability. The hip is flexed to 90 degrees, adducted, and internally rotated. Next, a posteriorly directed force is applied to the knee. When positive, this results in posterior pain or sensation of instability.⁷⁰

The LT test is performed with the hip flexed at 70 degrees and 30 degrees short of full abduction and the knee at 90 degrees of flexion. The hip is then maximally internally and externally rotated. Pain on either internal or external rotation is consistent with a positive test result. Pain should be relieved with rotation in the opposite direction and reproducible with rotation in the

direction of pain again. The sensitivity and specificity of the test are reported to be 90% and 85%, respectively.⁷¹

The resisted internal rotation test is one of the senior author's tests for gluteus medius and minimus pathology. The patient's hip and knee are each flexed to 90 degrees. In this position, the tip of the greater trochanter is directed posteriorly causing the much of the abductor complex to become an internal rotator of the hip. The clinician then instructs the patient to actively internally rotate the hip. This is often best conveyed by passively demonstrating which direction the patient will be moving their leg and foot. The clinician then resists the patient's internal rotation force. Pain and/or weakness at the lateral aspect of the hip is considered a positive test.

Lateral

Several examinations may be performed to help evaluate the hip in the lateral position. Flexing and extending the hip while moving the abductor mechanism across the hip can recreate visual ITB snapping. However, the patient will usually be able to actively reproduce a snapping ITB better than the clinician.

Abductor testing can be performed from the lateral position. Hip abduction against gravity or resistance may reveal weakness or pain. When weakness is discovered, the cause may be neurologic or musculotendinous. If neurologic weakness is suggested, an electromyogram can be a useful test for differentiating lumbar radiculopathy from superior gluteal nerve pathology. Painful weakness may suggest gluteus medius tendinopathy, trochanteric bursitis, or a trochanteric fracture. Further imaging with plain radiographs or MRI can be helpful in this setting.

Impingement testing with the flexion, adduction, and internal rotation maneuver can also be accomplished in the lateral position. While standing behind the patient, the clinician places one hand under the knee for support and the other hand on the hip joint with the index finger anterior and the thumb pointing posterior. If flexion, adduction, and internal rotation cause discomfort, the test is positive.

Ober testing comprises three parts: extension, neutral, and flexion. The extension test may detect an ITB contracture, the neutral test may detect a gluteus medius contracture, and the flexion test may detect a gluteus maximus contracture. The extension Ober test consists of flexing the knee, extending the hip, abducting the hip, and then releasing the hip. A delay in return to adduction indicates ITB pathology. The neutral Ober test consists of flexing the knee with the hip in the neutral position, abducting the hip, and then releasing the hip. A delay in return to adduction indicates gluteus medius pathology. The flexion Ober test consists of rotating the torso so that the patient's shoulders are flat on the table with the legs still in the lateral position, flexing the hip, extending the knee, abducting the hip, and then releasing the hip. A delay in return to adduction indicates gluteus maximus pathology.⁷²

Prone

The modified Thomas test and Ely test can differentiate between iliopsoas contracture and rectus femoris contracture. A positive modified Thomas test is simply an observation of the pelvis rising up in the prone position, which suggests an iliopsoas

contracture.⁶¹ The Ely test is performed by flexing the knee as far as possible with the patient prone. Upward motion of the buttocks and pelvis indicates a positive Ely test, which suggests a rectus femoris contracture because the rectus femoris crosses both the hip and knee joints.⁴⁵

The posterior impingement test is performed with the hip in extension and the leg flexed up. The hip is then externally rotated to bring the femoral head and neck into the posterior acetabulum. A positive test can be a sign of posterior labral pathology, excessive acetabular anteversion, or degenerative changes. Again, this test can assist the surgeon preoperatively and intraoperatively by indicating when to address a posterior lesion. While performing this maneuver, an anteriorly directed force can be placed on the posterior greater trochanter to translate the femoral head anteriorly. Pain or apprehension in the groin or anterior hip can be a sign of anterior instability.⁷⁴

The femoral nerve stretch test should also be conducted in this position by flexing the knee. Nerve type pain with this maneuver indicates femoral nerve pathology instead of hip joint pathology.

CONCLUSION

A thoughtful, systematic approach to the hip joint is necessary for the diagnosis and treatment of hip pathology. Because hip, back, and knee pain can easily be confused for one another, a consistent and careful history and physical examination are crucial to avoid misdiagnosis.⁴⁸ The physical examination maneuvers previously described as well as diagnostic injections should allow the clinician to narrow the differential diagnosis and help determine whether the patient's pain is intra-articular or extra-articular. These distinctions are important, given recent advances in imaging and treatment options for the hip.

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Buckland AJ, et al. Differentiating hip pathology from lumbar spine pathology: key points of evaluation and management. *J Am Acad Orthop Surg*. 2017;25(2):e23–e34.

Level of Evidence:

V

Summary:

This article discusses the difficulty in differentiating hip and spine pathology. Keys in the history, physical examination, diagnostic tests, and differential are discussed.

Citation:

Flynn JM, Widmann RF. The limping child: evaluation and diagnosis. *J Am Acad Orthop Surg*. 2001;9(2):89–98.

Level of Evidence:

V

Summary:

This article provides a framework for evaluating a child with a painful hip. The history, physical examination, and additional studies are outlined.

Citation:

Parvizi J, Leunig M, Ganz R. Femoroacetabular impingement. *J Am Acad Orthop Surg*. 2007;15(9):561–570.

Level of Evidence:

V

Summary:

In this review article on femoroacetabular impingement, the pathophysiology, history, physical examination, and treatment are discussed.

Citation:

Lynch TS, et al. Athletic hip injuries. *J Am Acad Orthop Surg*. 2017;25(4):269–279.

Level of Evidence:

V

Summary:

This review article discusses the “sports hip triad,” consisting of adductor strains, osteitis pubis, athletic pubalgia, or core muscle injury, often with underlying range-of-motion limitations secondary to femoroacetabular impingement.

Citation:

Domb BG, Brooks AG, Byrd JW. Clinical examination of the hip. *J Sports Rehab*. 2009;18:3–23.

Level of Evidence:

V

Summary:

This article provides a brief overview of the history, physical examination, and relevant imaging related to the athletic patient with hip pathology.

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